

```
[root@ashish-LFY-Article helloworld]# lsmod
Module                Size  Used by
ipmi_si                43275  1
ip6table_filter        2055  0
ip6_tables             19296  1 ip6table_filter
ebtable_nat            2039  0
ebtables              18401  1 ebtable_nat
ipt_MASQUERADE         2400  3
iptable_nat            6156  1
nf_nat                 23292  2 ipt_MASQUERADE,iptable_nat
nf_conntrack_ipv4      9440  4 iptable_nat,nf_nat
nf_defrag_ipv4        1449  1 nf_conntrack_ipv4
```

Figure 2: lsmod output

```
[root@ashish-LFY-Article helloworld]# insmod helloworld.ko
[root@ashish-LFY-Article helloworld]#
[root@ashish-LFY-Article helloworld]# lsmod | grep -i helloworld
helloworld            1360  0
[root@ashish-LFY-Article helloworld]#
[root@ashish-LFY-Article helloworld]# modinfo helloworld.ko
filename:
helloworld.ko
version: 0.3
description: Sample Module
author: ASHISH KUMAR
license: GPL
srcversion: 78C03435D3D0CFF5A9F9E
depends:
vermagic: 2.6.32-131.el5.el6.x86_64.debug SMP not_unload_no_modules
parm:
helli: int
[root@ashish-LFY-Article helloworld]#
[root@ashish-LFY-Article helloworld]# rmmod helloworld.ko
[root@ashish-LFY-Article helloworld]#
[root@ashish-LFY-Article helloworld]# lsmod | grep -i helloworld
[root@ashish-LFY-Article helloworld]#
```

Figure 3: Use of module management commands

```
/* This is the simple hello world kernel module program */
#include <linux/kernel.h> //Contains some often-used function prototypes
#include <linux/module.h> //Include all kernel modules
#include <linux/init.h> //Include the module_param() definition
#include <linux/moduleparam.h> //Include the module_param() definition

MODULE_LICENSE("GPL"); //Macro to specify the module license
MODULE_AUTHOR("ASHISH KUMAR"); //Macro to specify Module Author
MODULE_DESCRIPTION("Sample Module"); //Macro to give small description to module
MODULE_VERSION("0.1"); //Macro to give the module's version

static int hello = 2;
static int helo;

/*Module paramcount, int , &_R00000 //Macro to take routine arguments for module

static int hello_init(void) {
    for (int i = 1; i <= count; i++) {
        printk(KERN_ALERT "this is my first hello world program");
        printk(KERN_ALERT "the current running process in the system is : %s\n", (pid_t) 0);
    }
    return 0;
}

static void hello_exit(void) {
    printk(KERN_ALERT "goodbye we are done");
}

module_init(hello_init);
module_exit(hello_exit);
```

Figure 4: Code for a simple module

```
1 #BUIDIR: place holder for kernel build directory
2 BUIDIR := $(shell pwd)
3 ## Creating the module object file
4 obj-m := helloworld.o
5 ## Change the directory to BUIDIR and then build the module in the current working directory
6 default:
7 make -C $(BUIDIR) M=$(PWD) modules
8 ##
9 make -C $(BUIDIR) M=$(PWD) clean
```

Figure 5: A sample makefile

hardware from an application goes via the LKM to the kernel, and then to the actual hardware (see Figure 1).

To know the list of modules running in a Linux kernel you can use the 'lsmod' command, which actually gives the list of running modules at that point of time, by reading '/proc/modules' as shown in Figure 2.

Kernel modules can be broadly categorised as character, block or network modules.

Kernel module management commands

- insmod <module-name>:** This command is to insert the new module into the kernel
- lsmod:** This lists the modules that are currently loaded in the kernel

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